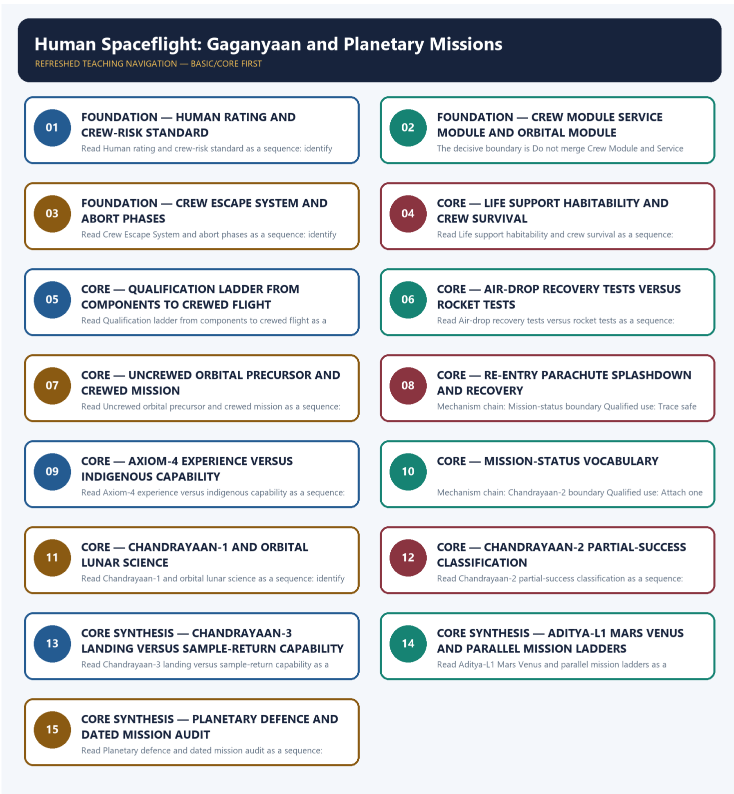
COMPLETE LEARNING SESSION

Human Spaceflight: Gaganyaan and Planetary Missions - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Human Spaceflight: Gaganyaan and Planetary Missions - Learner-v2 Complete Learning Session

> **Authoring-only generation:** 2026-09-03. No PDF was rendered and no tracker or index was mutated.

SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

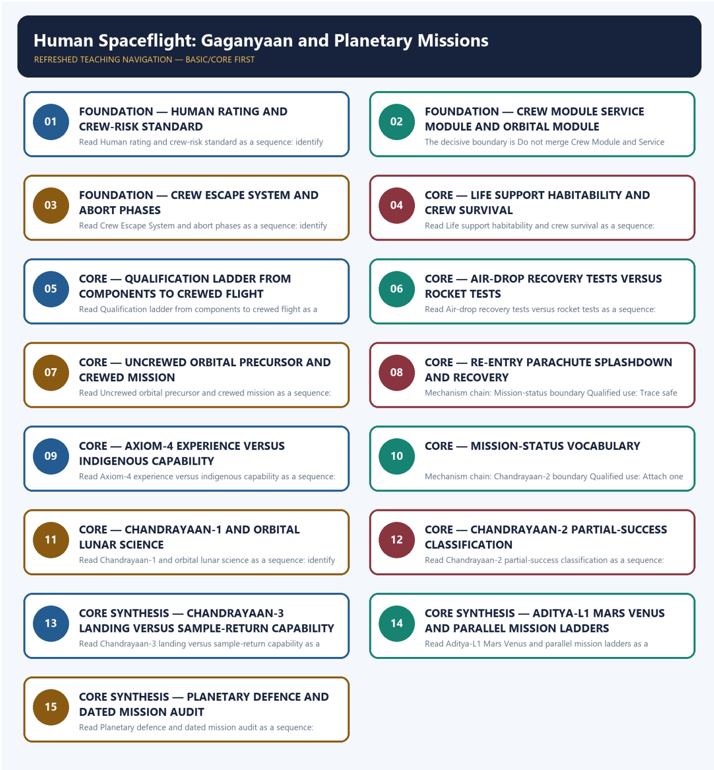
- **Generation date:** 2026-09-03.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. OCR-searchable official UPSC papers were supplementary only for printed routed demands. No answer key, payload capacity, mission result, constellation health, reactor output, isotope value or fusion milestone was inferred.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** Audited ledgers route the 2019 space-station, 2022 JWST, 2023 Chandrayaan-3 and 2024 asteroid Mains demands plus RTG, LISA and microgravity objective concepts here. No answer key, telescope specification or mission outcome is invented.
- **Live-link boundary:** The ISRO Gaganyaan page returned title-only on 2026-09-03. Repository owners remain the bounded source; crew, schedule, tests, payloads, findings, spacecraft health and mission status require a dated official update.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

LIVE OFFICIAL-SOURCE ATTEMPT LOG

The checks below were made on 2026-09-03. Substantive official text is used only for the proposition it supports; title-only, blocked or thin pages are recorded and supply no factual claim.

- https://www.isro.gov.in/Launchers.html - attempted 2026-09-03; direct retrieval returned only the page title, so no vehicle status, stage, propellant or payload-capacity claim was imported.
- https://www.isro.gov.in/Navigation.html - attempted 2026-09-03; direct retrieval returned only the page title, so no constellation, signal, coverage, accuracy or operational-status claim was imported.
- https://www.isro.gov.in/Gaganyaan.html - attempted 2026-09-03; direct retrieval returned only the page title, so no crew, schedule, test or mission-status claim was imported.

BASIC LEARNING SESSION



Refreshed teaching navigation

Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.

SESSION 1 - FOUNDATION - HUMAN RATING AND CREW-RISK STANDARD

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Technical definition: Read Human rating and crew-risk standard as a sequence: identify Human-rating boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Human rating**
- **crew-risk standard**
- **Mechanism chain**
- **Qualified use**
- **Human**

How to use them: Frame the answer through FOUNDATION; define Human rating, connect crew-risk standard with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
HUMAN RATING AND CREW-RISK STANDARD
01. Human-rating boundary
    |
    v
02. Orbital-module boundary
STATUS / CATEGORY FIREWALL -> Do not infer human rating from launch reliability alone.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

SIMPLE SYSTEM EXAMPLE

Read Human rating and crew-risk standard as a sequence: identify Human-rating boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not infer human rating from launch reliability alone. This keeps Human-rating boundary and Orbital-module boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

EXAMINER CAUTION

- Do not infer human rating from launch reliability alone.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Open with probability-of-crew-loss logic rather than prestige.

MINI RECAP

- **Mechanism chain:** Human-rating boundary -> Orbital-module boundary
- **Qualified use:** Open with probability-of-crew-loss logic rather than prestige.

CLOSING RECALL FLOW - FOUNDATION - HUMAN RATING AND CREW-RISK STANDARD

```
START / CONCEPT: FOUNDATION - Human rating and crew-risk standard
        |
        v
EXACT TERMS: FOUNDATION · Human rating · crew-risk standard · Mechanism chain · Qualified use ·
Human
        |
        v
MECHANISM / ARGUMENT: Read Human rating and crew-risk standard as a sequence: identify Human-rating
boundary, follow the named mechanism to its user or output, and stop at the last verified status
rather than assuming the next stage.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not infer human rating from launch reliability
alone.
        |
        v
UPSC TRAP / ANSWER-USE: Do not infer human rating from launch reliability alone.
        |
        v
ANSWER-GRABBING FORMULATION: Human rating is a system-level qualification of launcher margins,
redundancy, health monitoring, failure response and abort capability; an ordinary launch record
does not establish crew safety.
```

SESSION 2 - FOUNDATION - CREW MODULE SERVICE MODULE AND ORBITAL MODULE

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Technical definition: The decisive boundary is Do not merge Crew Module and Service Module functions.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Crew Module Service Module**
- **Orbital Module**
- **Mechanism chain**
- **Qualified use**
- **The Crew Escape System**

How to use them: Frame the answer through FOUNDATION; define Crew Module Service Module, connect Orbital Module with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
CREW MODULE SERVICE MODULE AND ORBITAL MODULE
01. Crew-escape boundary
STATUS / CATEGORY FIREWALL -> Do not merge Crew Module and Service Module functions.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

SIMPLE SYSTEM EXAMPLE

Read Crew Module Service Module and Orbital Module as a sequence: identify Crew-escape boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge Crew Module and Service Module functions. This keeps Crew-escape boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

EXAMINER CAUTION

- Do not merge Crew Module and Service Module functions.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Map habitability, re-entry, propulsion and power functions.

MINI RECAP

- **Mechanism chain:** Crew-escape boundary

- **Qualified use:** Map habitability, re-entry, propulsion and power functions.

CLOSING RECALL FLOW - FOUNDATION - CREW MODULE SERVICE MODULE AND ORBITAL MODULE

```
START / CONCEPT: FOUNDATION - Crew Module Service Module and Orbital Module
        |
        v
EXACT TERMS: FOUNDATION · Crew Module Service Module · Orbital Module · Mechanism chain · Qualified
use · The Crew Escape System
        |
        v
MECHANISM / ARGUMENT: Read Crew Module Service Module and Orbital Module as a sequence: identify
Crew-escape boundary, follow the named mechanism to its user or output, and stop at the last
verified status rather than assuming the next stage.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not merge Crew Module and Service Module
functions.
        |
        v
UPSC TRAP / ANSWER-USE: Do not merge Crew Module and Service Module functions.
        |
        v
ANSWER-GRABBING FORMULATION: The Crew Escape System is designed to pull the Crew Module away from a
failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not
a full orbital mission.
```

SESSION 3 - FOUNDATION - CREW ESCAPE SYSTEM AND ABORT PHASES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not call an abort test an orbital mission.

Technical definition: Read Crew Escape System and abort phases as a sequence: identify Life-support boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Crew Escape System and abort phases as a sequence: identify Life-support boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Crew Escape System**
- **abort phases**
- **Mechanism chain**
- **Qualified use**
- **Environmental Control**

How to use them: Frame the answer through FOUNDATION; define Crew Escape System, connect abort phases with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
CREW ESCAPE SYSTEM AND ABORT PHASES
01. Life-support boundary
STATUS / CATEGORY FIREWALL -> Do not call an abort test an orbital mission.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

SIMPLE SYSTEM EXAMPLE

Read Crew Escape System and abort phases as a sequence: identify Life-support boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call an abort test an orbital mission. This keeps Life-support boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

EXAMINER CAUTION

- Do not call an abort test an orbital mission.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Explain escape as a phase-specific survivability system.

MINI RECAP

- **Mechanism chain:** Life-support boundary
- **Qualified use:** Explain escape as a phase-specific survivability system.

CLOSING RECALL FLOW - FOUNDATION - CREW ESCAPE SYSTEM AND ABORT PHASES

```
START / CONCEPT: FOUNDATION - Crew Escape System and abort phases
        |
        v
EXACT TERMS: FOUNDATION · Crew Escape System · abort phases · Mechanism chain · Qualified use ·
Environmental Control
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Life-support boundary Qualified use: Explain escape as a
phase-specific survivability system.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not call an abort test an orbital mission.
        |
        v
UPSC TRAP / ANSWER-USE: Do not call an abort test an orbital mission.
        |
        v
ANSWER-GRABBING FORMULATION: Read Crew Escape System and abort phases as a sequence: identify Life-
support boundary, follow the named mechanism to its user or output, and stop at the last verified
status rather than assuming the next stage.
```

SESSION 4 - CORE - LIFE SUPPORT HABITABILITY AND CREW SURVIVAL

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not omit life support from human-spaceflight architecture.

Technical definition: Read Life support habitability and crew survival as a sequence: identify Qualification-ladder boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Life support habitability and crew survival as a sequence: identify Qualification-ladder boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Life support habitability**
- **crew survival**
- **Mechanism chain**
- **Qualified use**
- **Component**
- **Read Life**

How to use them: Frame the answer through Life support habitability; define crew survival, connect Mechanism chain with Qualified use to explain the mechanism, and use Component for the decisive comparison or qualification.

VISUAL FIRST

```
LIFE SUPPORT HABITABILITY AND CREW SURVIVAL
01. Qualification-ladder boundary
STATUS / CATEGORY FIREWALL -> Do not omit life support from human-spaceflight architecture.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

SIMPLE SYSTEM EXAMPLE

Read Life support habitability and crew survival as a sequence: identify Qualification-ladder boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not omit life support from human-spaceflight architecture. This keeps Qualification-ladder boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

EXAMINER CAUTION

- Do not omit life support from human-spaceflight architecture.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Identify atmosphere, pressure, thermal and carbon-dioxide control.

MINI RECAP

- **Mechanism chain:** Qualification-ladder boundary

- **Qualified use:** Identify atmosphere, pressure, thermal and carbon-dioxide control.

CLOSING RECALL FLOW - CORE - LIFE SUPPORT HABITABILITY AND CREW SURVIVAL

```
START / CONCEPT: CORE - Life support habitability and crew survival
        |
        v
EXACT TERMS: Life support habitability · crew survival · Mechanism chain · Qualified use ·
Component · Read Life
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Qualification-ladder boundary Qualified use: Identify
atmosphere, pressure, thermal and carbon-dioxide control.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not omit life support from human-spaceflight
architecture.
        |
        v
UPSC TRAP / ANSWER-USE: Do not omit life support from human-spaceflight architecture.
        |
        v
ANSWER-GRABBING FORMULATION: Read Life support habitability and crew survival as a sequence:
identify Qualification-ladder boundary, follow the named mechanism to its user or output, and stop
at the last verified status rather than assuming the next stage.
```

SESSION 5 - CORE - QUALIFICATION LADDER FROM COMPONENTS TO CREWED FLIGHT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not merge component, air-drop, abort and orbital tests.

Technical definition: Read Qualification ladder from components to crewed flight as a sequence: identify Air-drop-test boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Qualification ladder from components to crewed flight as a sequence: identify Air-drop-test boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Qualification ladder from components to crewed flight**
- **Mechanism chain**
- **Qualified use**
- **An Integrated Air Drop Test**
- **Read Qualification**
- **Air-drop-test**

How to use them: Frame the answer through Qualification ladder from components to crewed flight; define Mechanism chain, connect Qualified use with An Integrated Air Drop Test to explain the mechanism, and use Read Qualification for the decisive comparison or qualification.

VISUAL FIRST

```
QUALIFICATION LADDER FROM COMPONENTS TO CREWED FLIGHT
01. Air-drop-test boundary
STATUS / CATEGORY FIREWALL -> Do not merge component, air-drop, abort and orbital tests.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved.

SIMPLE SYSTEM EXAMPLE

Read Qualification ladder from components to crewed flight as a sequence: identify Air-drop-test boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge component, air-drop, abort and orbital tests. This keeps Air-drop-test boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved.

EXAMINER CAUTION

- Do not merge component, air-drop, abort and orbital tests.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Order every test rung and state what it validates.

MINI RECAP

- **Mechanism chain:** Air-drop-test boundary
- **Qualified use:** Order every test rung and state what it validates.

CLOSING RECALL FLOW - CORE - QUALIFICATION LADDER FROM COMPONENTS TO CREWED FLIGHT

```
START / CONCEPT: CORE - Qualification ladder from components to crewed flight
        |
        v
EXACT TERMS: Qualification ladder from components to crewed flight · Mechanism chain · Qualified
use · An Integrated Air Drop Test · Read Qualification · Air-drop-test
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Air-drop-test boundary Qualified use: Order every test rung
and state what it validates.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not merge component, air-drop, abort and
orbital tests.
        |
        v
UPSC TRAP / ANSWER-USE: Do not merge component, air-drop, abort and orbital tests.
        |
        v
ANSWER-GRABBING FORMULATION: Read Qualification ladder from components to crewed flight as a
sequence: identify Air-drop-test boundary, follow the named mechanism to its user or output, and
stop at the last verified status rather than assuming the next stage.
```

SESSION 6 - CORE - AIR-DROP RECOVERY TESTS VERSUS ROCKET TESTS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not call an air-drop test a rocket flight.

Technical definition: Read Air-drop recovery tests versus rocket tests as a sequence: identify Uncrewed-crewed boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Air-drop recovery tests versus rocket tests as a sequence: identify Uncrewed-crewed boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Air-drop recovery tests versus rocket tests**
- **Mechanism chain**
- **Qualified use**
- **Gaganyaan**
- **Read Air-drop**
- **Uncrewed-crewed**

How to use them: Frame the answer through Air-drop recovery tests versus rocket tests; define Mechanism chain, connect Qualified use with Gaganyaan to explain the mechanism, and use Read Air-drop for the decisive comparison or qualification.

VISUAL FIRST

```
AIR-DROP RECOVERY TESTS VERSUS ROCKET TESTS
01. Uncrewed-crewed boundary
    |
    v
02. Recovery-chain boundary
STATUS / CATEGORY FIREWALL -> Do not call an air-drop test a rocket flight.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

SIMPLE SYSTEM EXAMPLE

Read Air-drop recovery tests versus rocket tests as a sequence: identify Uncrewed-crewed boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call an air-drop test a rocket flight. This keeps Uncrewed-crewed boundary and Recovery-chain boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

EXAMINER CAUTION

- Do not call an air-drop test a rocket flight.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Separate parachute recovery testing from orbital flight.

MINI RECAP

- **Mechanism chain:** Uncrewed-crewed boundary -> Recovery-chain boundary
- **Qualified use:** Separate parachute recovery testing from orbital flight.

CLOSING RECALL FLOW - CORE - AIR-DROP RECOVERY TESTS VERSUS ROCKET TESTS

```
START / CONCEPT: CORE - Air-drop recovery tests versus rocket tests
        |
        v
EXACT TERMS: Air-drop recovery tests versus rocket tests · Mechanism chain · Qualified use ·
Gaganyaan · Read Air-drop · Uncrewed-crewed
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Uncrewed-crewed boundary - Recovery-chain boundary Qualified
use: Separate parachute recovery testing from orbital flight.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not call an air-drop test a rocket flight.
        |
        v
UPSC TRAP / ANSWER-USE: Do not call an air-drop test a rocket flight.
        |
        v
ANSWER-GRABBING FORMULATION: Read Air-drop recovery tests versus rocket tests as a sequence:
identify Uncrewed-crewed boundary, follow the named mechanism to its user or output, and stop at
the last verified status rather than assuming the next stage.
```

SESSION 7 - CORE - UNCREWED ORBITAL PRECURSOR AND CREWED MISSION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not rewrite a target date as an achieved mission.

Technical definition: Read Uncrewed orbital precursor and crewed mission as a sequence: identify Axiom-Gaganyaan boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

MUST-WRITE KEYWORDS

- **Uncrewed orbital precursor**
- **crewed mission**
- **Mechanism chain**
- **Qualified use**
- **Axiom**
- **Indian**

How to use them: Frame the answer through Uncrewed orbital precursor; define crewed mission, connect Mechanism chain with Qualified use to explain the mechanism, and use Axiom for the decisive comparison or qualification.

VISUAL FIRST

```
UNCREWED ORBITAL PRECURSOR AND CREWED MISSION
01. Axiom-Gaganyaan boundary
STATUS / CATEGORY FIREWALL -> Do not rewrite a target date as an achieved mission.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

SIMPLE SYSTEM EXAMPLE

Read Uncrewed orbital precursor and crewed mission as a sequence: identify Axiom-Gaganyaan boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not rewrite a target date as an achieved mission. This keeps Axiom-Gaganyaan boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

EXAMINER CAUTION

- Do not rewrite a target date as an achieved mission.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Place uncrewed integration before any crewed capability claim.

MINI RECAP

- **Mechanism chain:** Axiom-Gaganyaan boundary
- **Qualified use:** Place uncrewed integration before any crewed capability claim.

CLOSING RECALL FLOW - CORE - UNCREWED ORBITAL PRECURSOR AND CREWED MISSION

```
START / CONCEPT: CORE - Uncrewed orbital precursor and crewed mission
        |
        v
EXACT TERMS: Uncrewed orbital precursor · crewed mission · Mechanism chain · Qualified use · Axiom
· Indian
        |
        v
MECHANISM / ARGUMENT: Read Uncrewed orbital precursor and crewed mission as a sequence: identify
Axiom-Gaganyaan boundary, follow the named mechanism to its user or output, and stop at the last
verified status rather than assuming the next stage.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not rewrite a target date as an achieved
mission.
        |
        v
UPSC TRAP / ANSWER-USE: Do not rewrite a target date as an achieved mission.
        |
        v
ANSWER-GRABBING FORMULATION: Axiom-4 gave an Indian astronaut experience on an international ISS
mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous
crewed launch, orbital module or recovery capability.
```

SESSION 8 - CORE - RE-ENTRY PARACHUTE SPLASHDOWN AND RECOVERY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Re-entry parachute splashdown and recovery as a sequence: identify Mission-status boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: Mission-status boundary Qualified use: Trace safe return through re-entry, descent, splashdown and extraction.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Re-entry parachute splashdown and recovery as a sequence: identify Mission-status boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Re-entry parachute splashdown**
- **recovery**
- **Mechanism chain**
- **Qualified use**
- **Announced**
- **Cabinet-approved**

How to use them: Frame the answer through Re-entry parachute splashdown; define recovery, connect Mechanism chain with Qualified use to explain the mechanism, and use Announced for the decisive comparison or qualification.

VISUAL FIRST

```
RE-ENTRY PARACHUTE SPLASHDOWN AND RECOVERY
01. Mission-status boundary
STATUS / CATEGORY FIREWALL -> Do not stop the human-spaceflight chain at launch.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

SIMPLE SYSTEM EXAMPLE

Read Re-entry parachute splashdown and recovery as a sequence: identify Mission-status boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not stop the human-spaceflight chain at launch. This keeps Mission-status boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

EXAMINER CAUTION

- Do not stop the human-spaceflight chain at launch.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Trace safe return through re-entry, descent, splashdown and extraction.

MINI RECAP

- **Mechanism chain:** Mission-status boundary
- **Qualified use:** Trace safe return through re-entry, descent, splashdown and extraction.

CLOSING RECALL FLOW - CORE - RE-ENTRY PARACHUTE SPLASHDOWN AND RECOVERY

```
START / CONCEPT: CORE - Re-entry parachute splashdown and recovery
        |
        v
EXACT TERMS: Re-entry parachute splashdown · recovery · Mechanism chain · Qualified use · Announced
· Cabinet-approved
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Mission-status boundary Qualified use: Trace safe return
through re-entry, descent, splashdown and extraction.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not stop the human-spaceflight chain at launch.
        |
        v
UPSC TRAP / ANSWER-USE: Do not stop the human-spaceflight chain at launch.
        |
        v
ANSWER-GRABBING FORMULATION: Read Re-entry parachute splashdown and recovery as a sequence:
identify Mission-status boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 9 - CORE - AXIOM-4 EXPERIENCE VERSUS INDIGENOUS CAPABILITY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not call Axiom-4 an indigenous Gaganyaan flight.

Technical definition: Read Axiom-4 experience versus indigenous capability as a sequence: identify Chandrayaan-1 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Axiom-4 experience versus indigenous capability as a sequence: identify Chandrayaan-1 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Axiom-4 experience versus indigenous capability**
- **Mechanism chain**
- **Qualified use**
- **Chandrayaan**
- **Read Axiom**
- **The decisive boundary**

How to use them: Frame the answer through Axiom-4 experience versus indigenous capability; define Mechanism chain, connect Qualified use with Chandrayaan to explain the mechanism, and use Read Axiom for the decisive comparison or qualification.

VISUAL FIRST

```
AXIOM-4 EXPERIENCE VERSUS INDIGENOUS CAPABILITY
01. Chandrayaan-1 boundary
STATUS / CATEGORY FIREWALL -> Do not call Axiom-4 an indigenous Gaganyaan flight.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

SIMPLE SYSTEM EXAMPLE

Read Axiom-4 experience versus indigenous capability as a sequence: identify Chandrayaan-1 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call Axiom-4 an indigenous Gaganyaan flight. This keeps Chandrayaan-1 boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

EXAMINER CAUTION

- Do not call Axiom-4 an indigenous Gaganyaan flight.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Use international exposure as experience, not sovereign capability.

MINI RECAP

- **Mechanism chain:** Chandrayaan-1 boundary
- **Qualified use:** Use international exposure as experience, not sovereign capability.

CLOSING RECALL FLOW - CORE - AXIOM-4 EXPERIENCE VERSUS INDIGENOUS CAPABILITY

```
START / CONCEPT: CORE - Axiom-4 experience versus indigenous capability
        |
        v
EXACT TERMS: Axiom-4 experience versus indigenous capability · Mechanism chain · Qualified use ·
Chandrayaan · Read Axiom · The decisive boundary
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Chandrayaan-1 boundary Qualified use: Use international
exposure as experience, not sovereign capability.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not call Axiom-4 an indigenous Gaganyaan flight.
        |
        v
UPSC TRAP / ANSWER-USE: Do not call Axiom-4 an indigenous Gaganyaan flight.
        |
        v
ANSWER-GRABBING FORMULATION: Read Axiom-4 experience versus indigenous capability as a sequence:
identify Chandrayaan-1 boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 10 - CORE - MISSION-STATUS VOCABULARY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Mission-status vocabulary as a sequence: identify Chandrayaan-2 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: Chandrayaan-2 boundary Qualified use: Attach one exact status verb and date to every mission.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Mission-status vocabulary as a sequence: identify Chandrayaan-2 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Mission-status vocabulary**
- **Mechanism chain**
- **Qualified use**
- **Chandrayaan**
- **Read Mission-status**
- **The decisive boundary**

How to use them: Frame the answer through Mission-status vocabulary; define Mechanism chain, connect Qualified use with Chandrayaan to explain the mechanism, and use Read Mission-status for the decisive comparison or qualification.

VISUAL FIRST

```
MISSION-STATUS VOCABULARY
01. Chandrayaan-2 boundary
STATUS / CATEGORY FIREWALL -> Do not merge approved, operational, completed and concluded.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

SIMPLE SYSTEM EXAMPLE

Read Mission-status vocabulary as a sequence: identify Chandrayaan-2 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge approved, operational, completed and concluded. This keeps Chandrayaan-2 boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

EXAMINER CAUTION

- Do not merge approved, operational, completed and concluded.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Attach one exact status verb and date to every mission.

MINI RECAP

- **Mechanism chain:** Chandrayaan-2 boundary

- **Qualified use:** Attach one exact status verb and date to every mission.

CLOSING RECALL FLOW - CORE - MISSION-STATUS VOCABULARY

```
START / CONCEPT: CORE - Mission-status vocabulary
        |
        v
EXACT TERMS: Mission-status vocabulary · Mechanism chain · Qualified use · Chandrayaan · Read
Mission-status · The decisive boundary
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Chandrayaan-2 boundary Qualified use: Attach one exact
status verb and date to every mission.
        |
        v
CONSEQUENCE / CONTRAST: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not
complete soft landing; the mission must not be labelled either total success or total failure.
        |
        v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not merge approved, operational, completed and
concluded.
        |
        v
ANSWER-GRABBING FORMULATION: Read Mission-status vocabulary as a sequence: identify Chandrayaan-2
boundary, follow the named mechanism to its user or output, and stop at the last verified status
rather than assuming the next stage.
```

SESSION 11 - CORE - CHANDRAYAAN-1 AND ORBITAL LUNAR SCIENCE

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

Technical definition: Read Chandrayaan-1 and orbital lunar science as a sequence: identify Chandrayaan-3 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

MUST-WRITE KEYWORDS

- **Chandrayaan-1**
- **orbital lunar science**
- **Mechanism chain**
- **Qualified use**
- **Chandrayaan**
- **Chandrayaan-4**

How to use them: Frame the answer through Chandrayaan-1; define orbital lunar science, connect Mechanism chain with Qualified use to explain the mechanism, and use Chandrayaan for the decisive comparison or qualification.

VISUAL FIRST

```
CHANDRAYAAN-1 AND ORBITAL LUNAR SCIENCE
01. Chandrayaan-3 boundary
    |
    v
02. Chandrayaan-4 boundary
STATUS / CATEGORY FIREWALL -> Do not equate lunar orbit science with soft landing.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

SIMPLE SYSTEM EXAMPLE

Read Chandrayaan-1 and orbital lunar science as a sequence: identify Chandrayaan-3 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not equate lunar orbit science with soft landing. This keeps Chandrayaan-3 boundary and Chandrayaan-4 boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.
- Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

EXAMINER CAUTION

- Do not equate lunar orbit science with soft landing.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Separate science discovery from landing capability.

MINI RECAP

- **Mechanism chain:** Chandrayaan-3 boundary -> Chandrayaan-4 boundary
- **Qualified use:** Separate science discovery from landing capability.

CLOSING RECALL FLOW - CORE - CHANDRAYAAN-1 AND ORBITAL LUNAR SCIENCE

```
START / CONCEPT: CORE - Chandrayaan-1 and orbital lunar science
        |
        v
EXACT TERMS: Chandrayaan-1 · orbital lunar science · Mechanism chain · Qualified use · Chandrayaan
· Chandrayaan-4
        |
        v
MECHANISM / ARGUMENT: Read Chandrayaan-1 and orbital lunar science as a sequence: identify
Chandrayaan-3 boundary, follow the named mechanism to its user or output, and stop at the last
verified status rather than assuming the next stage.
        |
        v
CONSEQUENCE / CONTRAST: Chandrayaan-3 demonstrated a successful soft landing and surface operations;
 that achievement does not by itself establish lunar ascent, docking or sample return.
        |
        v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not equate lunar orbit science with soft
landing.
        |
        v
ANSWER-GRABBING FORMULATION: Chandrayaan-4 is an approved lunar sample-return technology mission in
the repository owner; approval and intended completion are not launch, landing or returned material.
```

SESSION 12 - CORE - CHANDRAYAAN-2 PARTIAL-SUCCESS CLASSIFICATION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not call Chandrayaan-2 a total failure.

Technical definition: Read Chandrayaan-2 partial-success classification as a sequence: identify Aditya-L1 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The decisive boundary is Do not call Chandrayaan-2 a total failure.

MUST-WRITE KEYWORDS

- **Chandrayaan-2 partial-success classification**
- **Mechanism chain**
- **Qualified use**
- **Aditya-L1**
- **Aditya**
- **Sun-Earth**

How to use them: Frame the answer through Chandrayaan-2 partial-success classification; define Mechanism chain, connect Qualified use with Aditya-L1 to explain the mechanism, and use Aditya for the decisive comparison or qualification.

VISUAL FIRST

```
CHANDRAYAAN-2 PARTIAL-SUCCESS CLASSIFICATION
01. Aditya-L1 boundary
STATUS / CATEGORY FIREWALL -> Do not call Chandrayaan-2 a total failure.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

SIMPLE SYSTEM EXAMPLE

Read Chandrayaan-2 partial-success classification as a sequence: identify Aditya-L1 boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call Chandrayaan-2 a total failure. This keeps Aditya-L1 boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

EXAMINER CAUTION

- Do not call Chandrayaan-2 a total failure.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Write orbiter success and lander failure in the same sentence.

MINI RECAP

- **Mechanism chain:** Aditya-L1 boundary

- **Qualified use:** Write orbiter success and lander failure in the same sentence.

CLOSING RECALL FLOW - CORE - CHANDRAYAAN-2 PARTIAL-SUCCESS CLASSIFICATION

```
START / CONCEPT: CORE - Chandrayaan-2 partial-success classification
        |
        v
EXACT TERMS: Chandrayaan-2 partial-success classification · Mechanism chain · Qualified use ·
Aditya-L1 · Aditya · Sun-Earth
        |
        v
MECHANISM / ARGUMENT: Read Chandrayaan-2 partial-success classification as a sequence: identify
Aditya-L1 boundary, follow the named mechanism to its user or output, and stop at the last verified
status rather than assuming the next stage.
        |
        v
CONSEQUENCE / CONTRAST: Do not call Chandrayaan-2 a total failure.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: write orbiter success and lander
failure in the same sentence.
        |
        v
ANSWER-GRABBING FORMULATION: The decisive boundary is Do not call Chandrayaan-2 a total failure.
```

SESSION 13 - CORE SYNTHESIS - CHANDRAYAAN-3 LANDING VERSUS SAMPLE-RETURN CAPABILITY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not infer sample return from Chandrayaan-3.

Technical definition: Read Chandrayaan-3 landing versus sample-return capability as a sequence: identify Lagrange-station-keeping boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Chandrayaan-3 landing versus sample-return capability as a sequence: identify Lagrange-station-keeping boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **Chandrayaan-3 landing versus sample-return capability**
- **Mechanism chain**
- **Qualified use**
- **Lagrange-point**
- **Read Chandrayaan**

How to use them: Frame the answer through CORE SYNTHESIS; define Chandrayaan-3 landing versus sample-return capability, connect Mechanism chain with Qualified use to explain the mechanism, and use Lagrange-point for the decisive comparison or qualification.

VISUAL FIRST

```
CHANDRAYAAN-3 LANDING VERSUS SAMPLE-RETURN CAPABILITY
01. Lagrange-station-keeping boundary
STATUS / CATEGORY FIREWALL -> Do not infer sample return from Chandrayaan-3.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point.

SIMPLE SYSTEM EXAMPLE

Read Chandrayaan-3 landing versus sample-return capability as a sequence: identify Lagrange-station-keeping boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not infer sample return from Chandrayaan-3. This keeps Lagrange-station-keeping boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point.

EXAMINER CAUTION

- Do not infer sample return from Chandrayaan-3.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Separate landing, ascent, docking and Earth-return capabilities.

MINI RECAP

- **Mechanism chain:** Lagrange-station-keeping boundary
- **Qualified use:** Separate landing, ascent, docking and Earth-return capabilities.

CLOSING RECALL FLOW - CORE SYNTHESIS - CHANDRAYAAN-3 LANDING VERSUS SAMPLE-RETURN CAPABILITY

```
START / CONCEPT: CORE SYNTHESIS - Chandrayaan-3 landing versus sample-return capability
        |
        v
EXACT TERMS: CORE SYNTHESIS · Chandrayaan-3 landing versus sample-return capability · Mechanism
chain · Qualified use · Lagrange-point · Read Chandrayaan
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Lagrange-station-keeping boundary Qualified use: Separate
landing, ascent, docking and Earth-return capabilities.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not infer sample return from Chandrayaan-3.
        |
        v
UPSC TRAP / ANSWER-USE: A halo orbit around L1 provides sustained solar viewing but requires
station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at
a force-free point.
        |
        v
ANSWER-GRABBING FORMULATION: Read Chandrayaan-3 landing versus sample-return capability as a
sequence: identify Lagrange-station-keeping boundary, follow the named mechanism to its user or
output, and stop at the last verified status rather than assuming the next stage.
```

SESSION 14 - CORE SYNTHESIS - ADITYA-L1 MARS VENUS AND PARALLEL MISSION LADDERS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Technical definition: Read Aditya-L1 Mars Venus and parallel mission ladders as a sequence: identify Mars-Venus boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Aditya-L1 Mars Venus and parallel mission ladders as a sequence: identify Mars-Venus boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **Aditya-L1 Mars Venus**
- **parallel mission ladders**
- **Mechanism chain**
- **Qualified use**
- **Mars Orbiter Mission**

How to use them: Frame the answer through CORE SYNTHESIS; define Aditya-L1 Mars Venus, connect parallel mission ladders with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
ADITYA-L1 MARS VENUS AND PARALLEL MISSION LADDERS
01. Mars-Venus boundary
    |
    v
02. Capability-ladder boundary
STATUS / CATEGORY FIREWALL -> Do not treat Chandrayaan-4 approval as returned samples.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

SIMPLE SYSTEM EXAMPLE

Read Aditya-L1 Mars Venus and parallel mission ladders as a sequence: identify Mars-Venus boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not treat Chandrayaan-4 approval as returned samples. This keeps Mars-Venus boundary and Capability-ladder boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.
- Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

EXAMINER CAUTION

- Do not treat Chandrayaan-4 approval as returned samples.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Use distinct ladders for solar, lunar and planetary missions.

MINI RECAP

- **Mechanism chain:** Mars-Venus boundary -> Capability-ladder boundary
- **Qualified use:** Use distinct ladders for solar, lunar and planetary missions.

CLOSING RECALL FLOW - CORE SYNTHESIS - ADITYA-L1 MARS VENUS AND PARALLEL MISSION LADDERS

```
START / CONCEPT: CORE SYNTHESIS - Aditya-L1 Mars Venus and parallel mission ladders
        |
        v
EXACT TERMS: CORE SYNTHESIS · Aditya-L1 Mars Venus · parallel mission ladders · Mechanism chain ·
Qualified use · Mars Orbiter Mission
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Mars-Venus boundary - Capability-ladder boundary Qualified
use: Use distinct ladders for solar, lunar and planetary missions.
        |
        v
CONSEQUENCE / CONTRAST: Mars Orbiter Mission is a concluded historic mission, whereas the Venus
Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy
capability does not prove the future mission's outcome.
        |
        v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not treat Chandrayaan-4 approval as returned
samples.
        |
        v
ANSWER-GRABBING FORMULATION: Read Aditya-L1 Mars Venus and parallel mission ladders as a sequence:
identify Mars-Venus boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 15 - CORE SYNTHESIS - PLANETARY DEFENCE AND DATED MISSION AUDIT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Claim: Planetary defence is chiefly an early-warning problem -> NASA’s DART changed Dimorphos’s orbit through a kinetic-impact demonstration -> it shows a deflection mechanism can work with warning and characterization -> qualification: one controlled test is not a universal shield; undiscovered objects, object structure and short-warning cases constrain response.

Technical definition: Read Planetary defence and dated mission audit as a sequence: identify Planetary-defence boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Planetary defence and dated mission audit as a sequence: identify Planetary-defence boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **Planetary defence**
- **dated mission audit**
- **Mechanism chain**

- **Qualified use**
- **Subject**

How to use them: Frame the answer through CORE SYNTHESIS; define Planetary defence, connect dated mission audit with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
PLANETARY DEFENCE AND DATED MISSION AUDIT
01. Planetary-defence boundary
    |
    v
02. Volatile mission boundary
STATUS / CATEGORY FIREWALL -> Do not call Aditya-L1 a solar landing.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

SIMPLE SYSTEM EXAMPLE

Read Planetary defence and dated mission audit as a sequence: identify Planetary-defence boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call Aditya-L1 a solar landing. This keeps Planetary-defence boundary and Volatile mission boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.
- Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

EXAMINER CAUTION

- Do not call Aditya-L1 a solar landing.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Close with detect-characterise-warn-deflect and fresh official evidence.

MINI RECAP

- **Mechanism chain:** Planetary-defence boundary -> Volatile mission boundary
- **Qualified use:** Close with detect-characterise-warn-deflect and fresh official evidence.

COMPLETE BASIC OWNER EVIDENCE BANK

> **Subject:** Science & Technology | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III + GS-II (international cooperation) + Prelims. **Core area:** Human-spaceflight architecture, astronaut training and India's lunar/solar/planetary mission sequence. **Grounded in:** ISRO Gaganyaan page (https://www.isro.gov.in/Gaganyaan.html); ISRO Axiom-4 launch and conclusion notes (https://www.isro.gov.in/ISRO_EN/Successful_Launch_of_Axiom_Mission.html ; https://www.isro.gov.in/Axiom04_mission_successfully_concluded_return_ISRO_Gaganyatri_ShubhanshuShukla.html); Integrated Air Drop Tests (https://www.isro.gov.in/Integrated_Air_Drop_Test_for_Gaganyaan_Missions.html ; https://www.isro.gov.in/ISRO_conducts_Second_Integrated_Air_Drop_Test_for_Gaganyaan.html) and IMAT-05 (https://www.isro.gov.in/Integrated_Main_Parachute_Air_Drop_Test.html); Pad Abort Test note (https://pib.gov.in/PressReleasePage.aspx?PRID=1540066); ISRO Chandrayaan pages (https://www.isro.gov.in/Chandrayaan-1_science.html ; https://www.isro.gov.in/Chandrayaan2_science.html ; https://www.isro.gov.in/Chandrayaan3.html); ISRO Aditya-L1 pages (https://www.isro.gov.in/Aditya_L1.html ; https://www.isro.gov.in/ISRO_EN/halo-orbit-insertion-adtya-l1.html); Cabinet approvals of 18 Sep 2024 for Gaganyaan/BAS, Chandrayaan-4 and the Venus Orbiter Mission (https://pib.gov.in/PressReleasePage.aspx?PRID=2055978 ; https://pib.gov.in/PressReleasePage.aspx?PRID=2055989 ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=2055982); Gaganyaan status replies (https://pib.gov.in/PressReleasePage.aspx?PRID=2205291 ; https://pib.gov.in/PressReleasePage.aspx?PRID=2227019) - re-verified 2 Aug 2026. FACT: = source-grounded | ANALYSIS: = analytical inference | CURRENT: = current/dated development. *Companion:* `advanced/03_Human-Spaceflight-Gaganyaan-and-Planetary-Missions.md`.

1. Visual foundation

```
GAGANYAAN STACK
Human-rated LVM3
      |
      v
Orbital Module = Crew Module (habitable) + Service Module (support/propulsion)
      |
      v
Uncrewed tests / abort tests -> crewed mission -> splashdown recovery in Indian waters
```

Mission / programme	What UPSC should remember	Current broad status
Gaganyaan	Indigenous human spaceflight capability demonstration	Under development / uncrewed test phase (as of Aug 2026 no Gaganyaan orbital flight has taken place)
Chandrayaan-1	First lunar mission; major science outcome on lunar water/hydroxyl evidence	Completed
Chandrayaan-2	Orbiter successful; lander soft landing not achieved	Orbiter reported operational by ISRO; re-verify the orbiter's status before quoting it as "currently operating"
Chandrayaan-3	Successful soft landing (23 Aug 2023) and surface science near the lunar south polar region	Completed science mission
Aditya-L1	Solar observatory in a halo orbit around Sun-Earth L1 (insertion 6 Jan 2024)	Operational
Mars Orbiter Mission	Historic low-cost Mars orbiter	Mission concluded
Chandrayaan-4	Lunar **sample-return** technology demonstration	Approved 18 Sep 2024 / under development
Venus Orbiter Mission / Shukrayaan	Next major comparative-planetology step	Approved 18 Sep 2024; launch opportunity stated as March 2028
Bharatiya Antariksh Station (BAS)	Indian space station; BAS-01 is the first module	Approved 18 Sep 2024; BAS-01 targeted December 2028

MEMORY: **The four ladders - never mix them.** (a) *Human spaceflight*: test -> uncrewed orbital -> crewed -> station. (b) *Lunar*: orbit -> soft land -> rove -> sample return. (c) *Solar/deep-space observation*: Lagrange-point station-keeping, no landing at all. (d) *International crewed cooperation* (e.g. an ISS flight) - participation in someone else's system, not indigenous capability.

2. Essential definitions

Concept	Exam-ready meaning
FACT: **Human-rated launch vehicle**	Launcher modified and certified to meet stricter reliability and crew-safety requirements.
FACT: **Crew Module (CM)**	Habitable, re-entry capable part of Gaganyaan that carries astronauts in an Earth-like environment.
FACT: **Service Module (SM)**	Support module that provides propulsion, power and other non-habitable orbital support to the Crew Module.
FACT: **Orbital Module (OM)**	Combined Gaganyaan flight element made of the Crew Module and Service Module.
FACT: **Vyommitra**	ISRO's humanoid mission-support robot intended for uncrewed human-spaceflight precursor missions.
FACT: **Crew Escape System (CES)**	A set of solid motors that pulls the Crew Module clear of a failing launch vehicle. It must work from the launch pad right through the atmospheric ascent - this is what the Pad Abort Test (5 Jul 2018) and Test Vehicle mission TV-D1 (21 Oct 2023) demonstrated.
ANALYSIS: **Abort mode**	A pre-planned survivable exit from a failing flight, defined for each phase (pad abort, low-altitude, high-altitude, orbital). A launcher without validated abort modes cannot be human-rated regardless of its success record.
ANALYSIS: **Environmental Control and Life Support System (ECLSS)**	Onboard system maintaining breathable atmosphere, pressure, temperature, humidity and CO_2 removal - the defining difference between a satellite and a crewed spacecraft.
ANALYSIS: **Integrated Air Drop Test (IADT)**	Test in which a crew-module mass simulator is dropped from a helicopter/aircraft to qualify the **parachute-based deceleration and recovery sequence**, without any rocket launch. IADT-01 was conducted on 24 Aug 2025 and IADT-02 on 10 Apr 2026.
FACT: **Halo orbit**	A periodic three-dimensional orbit around a Lagrange point; Aditya-L1 uses one around Sun-Earth L1. A satellite there needs continuous **station-keeping** because the orbit is only quasi-stable.
ANALYSIS: **Lagrange point**	One of five positions in a two-body system where gravitational and centripetal effects allow a small body to hold a fixed configuration. L1 (between Sun and Earth) gives an uninterrupted view of the Sun without eclipses - the reason Aditya-L1 was placed there rather than in Earth orbit.
FACT: **Soft landing**	Controlled, non-destructive touchdown on a celestial body, with descent velocity reduced enough that the spacecraft survives intact. (Preserved payload function is the intended *result*; the defining condition is the controlled, survivable descent.)
FACT: **Sample return**	Mission design that collects extraterrestrial material and brings it back to Earth for analysis. It requires two capabilities India has not yet demonstrated together: ascent from another body, and rendezvous/docking or direct Earth return with containment.
FACT: **Gaganyatri**	Indian astronaut-designate/crew member associated with the Gaganyaan programme.

3. Mechanism / how it works

1. For Gaganyaan, ISRO must human-rate the launcher, qualify crew escape and life-support systems, test the orbital module and validate recovery procedures before crewed flight.
2. The orbital module consists of a Crew Module for astronauts and a Service Module for propulsion and support; safety is built through redundancy, abort capability and progressive testing.
3. Uncrewed and abort-test missions are essential because human-spaceflight credibility comes from demonstrated survivability, not just from launch success.
4. Planetary missions follow a different logic: launch, orbit-raising/transit, insertion into the destination environment and then science operations or landing/surface operations depending on mission type.
5. Lunar missions, solar observation, Mars exploration and approved Venus/sample-return plans together form a stepwise capability ladder in deep-space science and technology.

4. Institutions and programmes

- FACT: **ISRO / Human Space Flight Centre (HSFC):** leads the Gaganyaan programme and coordinates the human-spaceflight ecosystem across ISRO centres, labs, academia and industry.
- FACT: **Astronaut Training Facility, Bengaluru:** ISRO states that training covers academics, simulators, physical fitness, aero-medical work, survival and procedures.
- FACT: **Indian Navy and allied agencies:** support recovery architecture for splashdown missions.
- FACT: **ISRO science and mission centres:** execute lunar, solar and planetary missions through mission design, deep-space tracking and payload operations.
- FACT: **International cooperation channels:** Axiom-4/ISS participation shows that foreign mission exposure can complement, but does not replace, indigenous crewed capability development.
- FACT: **Cabinet-approved future architecture:** BAS-1, Chandrayaan-4 and Venus Orbiter Mission belong to the next phase of India's human-spaceflight and planetary roadmap.

5. Indian applications, examples and limitations

- FACT: Human-spaceflight drives life support, crew safety, re-entry, recovery, materials and avionics capability that can spill over into wider aerospace engineering.
- FACT: Chandrayaan missions deepen lunar science, landing technology and surface-operation know-how; Chandrayaan-3 especially turned landing success into a major technological milestone.
- FACT: Aditya-L1 enables continuous solar observation and space-weather science from halo orbit rather than from low Earth orbit.
- FACT: Axiom-4 participation gave an Indian astronaut operational exposure to the ISS ecosystem and microgravity experiments, adding practical experience to the broader human-spaceflight journey.
- ANALYSIS: Limitation: human-spaceflight timelines are highly sensitive to safety validation; delays are not automatically failures but often the cost of reliability.
- ANALYSIS: Limitation: deep-space missions require long-duration communication, navigation, thermal control and mission-operations maturity that cannot be built instantly.
- ANALYSIS: Limitation: prestige missions draw attention, but long-term value depends on sustained science output, industrial learning and follow-on missions.
- ANALYSIS: Limitation: approved future missions such as BAS or Venus Orbiter should never be written as if they are already operating in space.

6. Must-Know Facts for Prelims

- FACT: ISRO states that Gaganyaan envisages sending a crew of three members to about 400 km orbit for about three days and recovering them in Indian sea waters.
- FACT: ISRO identifies Human Rated LVM3 (HLVM3) as the launcher for Gaganyaan.
- FACT: The Gaganyaan orbital module consists of a Crew Module and a Service Module.
- FACT: Chandrayaan-1 science results on the ISRO site list detection of hydroxyl/water molecules on the lunar surface and evidence related to subsurface water-ice in permanently shadowed regions.

- FACT: Chandrayaan-2 was not a total failure: ISRO states the orbiter remains healthy and payloads are operational even though the lander did not complete soft landing.
- FACT: Chandrayaan-3 achieved successful soft landing and in-situ lunar science including sulphur detection on the surface.
- FACT: ISRO states Aditya-L1 entered halo orbit around Sun-Earth L1 on 6 January 2024.
- FACT: Mangalyaan outlived its original design life and is now treated in recent official references as a concluded mission.
- FACT: **Gaganyaan test sequence actually completed so far (all uncrewed):** Pad Abort Test (5 Jul 2018) -> Test Vehicle abort mission **TV-D1** (21 Oct 2023) -> **IADT-01** parachute/recovery air-drop test (24 Aug 2025) -> **IADT-02** (10 Apr 2026) -> Integrated Main Parachute Air Drop Test **IMAT-05** (7 Jul 2026, qualifying parachutes for the first uncrewed orbital mission G1).
- FACT: **G1, the first uncrewed Gaganyaan orbital flight, had not been flown as of the latest verified official statements (Aug 2026).** The first **crewed** flight is officially targeted in **2027-28**. Treat both as targets, not achievements.
- FACT: **Bharatiya Antariksh Station:** approved 18 Sep 2024 within an expanded Gaganyaan scope; **BAS-01 (first module) targeted by December 2028**.
- FACT: **Chandrayaan-4** (lunar sample return) and the **Venus Orbiter Mission** were both approved on **18 Sep 2024**; the Venus launch opportunity was stated as **March 2028**.
- FACT: **Axiom-4:** Group Captain **Shubhanshu Shukla** flew as mission pilot, launching **25 Jun 2025** and returning **15 Jul 2025** after an **18-day** mission, carrying out **seven Indian experiments** aboard the ISS. This was an **international ISS mission on a SpaceX Dragon**, not an Indian human spaceflight.

7. UPSC traps

- WRONG: Axiom-4 made Gaganyaan unnecessary. -> Axiom-4 gave international crewed-mission exposure; Gaganyaan is about indigenous launch, orbital and recovery capability.
- WRONG: Chandrayaan-2 was a complete failure. -> The lander did not soft-land successfully, but the orbiter was successfully placed in lunar orbit and continues science operations per ISRO.
- WRONG: Aditya-L1 means India has “landed on the Sun.” -> It is a solar observatory placed in halo orbit around Sun-Earth L1 for continuous observation.
- WRONG: Vyommitra is an Indian astronaut. -> Vyommitra is a humanoid mission-support robot for uncrewed test phases.
- WRONG: Mangalyaan should be described as operational today. -> Recent official references celebrate it as a concluded historic mission.
- WRONG: Approved Bharatiya Antariksh Station or Venus Orbiter plans should be written as completed missions. -> They are approved/under-development roadmap items, not accomplished missions.
- WRONG: "An air-drop test is a Gaganyaan flight." -> IADT/IMAT tests drop a crew-module simulator from an aircraft or helicopter to qualify parachutes and recovery; no launch vehicle is involved.
- WRONG: "TV-D1 was India's first uncrewed Gaganyaan mission." -> TV-D1 was a **Test Vehicle abort demonstration** validating the Crew Escape System in transonic flight; the first uncrewed *orbital* mission is G1, which is distinct and later.
- WRONG: Writing that an Indian astronaut has "gone to space under Gaganyaan." -> Shubhanshu Shukla flew on Axiom-4 to the ISS on a US commercial vehicle; Gaganyaan's crewed flight is targeted for 2027-28.

8. CURRENT: Current anchor

- CURRENT: **06 Jan 2024 | Aditya-L1 - operational in halo orbit.** ISRO said halo-orbit insertion at Sun-Earth L1 was successfully accomplished; use this to mark the mission as operational, not proposed.
- CURRENT: **18 Sep 2024 | Gaganyaan follow-on + Bharatiya Antariksh Station - approved.** Cabinet approved BAS-01 and follow-on missions within the expanded Gaganyaan scope; **BAS-01 target: December 2028.**
- CURRENT: **18 Sep 2024 | Venus Orbiter Mission and Chandrayaan-4 - approved.** Chandrayaan-4 was approved as a lunar **sample-return technology** mission (Cabinet wording: completion within 36 months); the Venus Orbiter Mission's stated opportunity is **March 2028**.

- CURRENT: **25 Jun - 15 Jul 2025 | Axiom Mission 4 - flown and concluded.** Group Captain Shubhanshu Shukla flew as mission pilot, spent 18 days on the mission and performed seven Indian experiments. **Status: completed international mission.**
- CURRENT: **24 Aug 2025 / 10 Apr 2026 | Integrated Air Drop Tests IADT-01 and IADT-02 - conducted.** Qualified the crew-module parachute-deceleration and recovery sequence.
- CURRENT: **7 Jul 2026 | IMAT-05 main-parachute air-drop test - conducted.** ISRO described it as a qualification step towards the first uncrewed orbital mission **G1**.
- CURRENT: **17 Dec 2025 / 12 Feb 2026 | Gaganyaan - crewed flight targeted 2027-28; G1 preparation continuing.** Parliamentary material described crew/service-module checks, integration and simulations. **Status: under development; no Gaganyaan orbital flight had occurred as of the latest verified statement.**

ANALYSIS: **Currentness note:** The dated statuses above are accurate to the cited source date (latest re-verification 2 Aug 2026); verify later updates before exam use.

10. Mains framework / angles

- ANALYSIS: Separate human-spaceflight capability from planetary-science capability, then show how both feed a larger national roadmap.
- ANALYSIS: Use a status vocabulary carefully: completed, operational, approved, under development, tested or concluded - vague wording loses marks.
- ANALYSIS: Bring in the sequence logic: Chandrayaan-3 -> Chandrayaan-4 approval; Gaganyaan -> BAS approval; Mars legacy -> Venus as next comparative-planetology step.
- ANALYSIS: International exposure (ISS/Axiom) should be framed as complementary experience, not as a substitute for indigenous capacity.

> **Answer thesis:** India’s human-spaceflight and planetary programme is a staged capability-building project in which crew safety, orbital systems, landing technologies, deep-space operations and international mission exposure reinforce one another; the key UPSC skill is to distinguish clearly between completed successes, partial successes, operational missions and merely approved future ambitions.

11. Probable questions

- ANALYSIS: **Prelims (practice):** Which of the following correctly matches the mission with its present status: Chandrayaan-2 orbiter, Aditya-L1, Mangalyaan and Venus Orbiter Mission?
- ANALYSIS: **Mains (10 marks, practice):** Why should Axiom-4 participation be analysed as complementary to, but not a substitute for, Gaganyaan? Answer in 150 words.
- ANALYSIS: **Mains (15 marks, practice):** Discuss how Gaganyaan, Bharatiya Antariksh Station plans and India’s lunar/solar/planetary missions together indicate a shift from episodic missions to a long-horizon space capability strategy.

12. Study links

- FACT: Advanced companion: `advanced/03_Human-Spaceflight-Gaganyaan-and-Planetary-Missions.md`.
- FACT: `01_Space-Programme-ISRO-Launch-Vehicles.md` - launcher capability, human-rating and space-governance context.
- FACT: `02_Satellites-NavIC-GAGAN-and-Applications.md` - mission applications and space infrastructure base.
- FACT: `16_Nanotechnology-and-Applications.md` - materials, sensors and avionics relevance to space systems.

Core answer architecture - human spaceflight, planetary capability and planetary defence

Thesis choice. Treat crewed flight, planetary science and planetary defence as adjacent but different capability ladders: one test or foreign mission cannot certify all three.

10-mark spine. Identify the mission class; show its minimum mechanism/safety requirement; name one Indian achievement or approved next step; then date-stamp the exact status before judging value.

15/20-mark spine. Use **mission objective -> technical chain -> named evidence -> scientific/strategic spillover -> safety, cost and maturity limits**. For asteroids, use **hazard assessment -> detection/tracking -> deflection options -> residual uncertainty**, rather than a generic space-programme paragraph.

Evidence units.

- **Claim:** Indigenous human-spaceflight is a safety system, not a launch event -> **HLVM3, Crew/Service Module, Crew Escape System and air-drop/parachute qualification steps** -> each validates a survivability layer needed before crew risk is accepted -> **qualification:** abort/parachute tests and an international ISS flight are not a Gaganyaan orbital or crewed mission.
- **Claim:** India's planetary programme compounds capabilities over missions -> **Chandrayaan-3 soft landing, Aditya-L1 halo-orbit operation, and approved Chandrayaan-4/Venus Orbiter plans** -> landing, deep-space operations and sample-return preparation create a longer technical ladder -> **qualification:** approval/target dates do not establish a launched or operating future mission.
- **Claim:** Planetary defence is chiefly an early-warning problem -> **NASA's DART changed Dimorphos's orbit through a kinetic-impact demonstration** -> it shows a deflection mechanism can work with warning and characterization -> **qualification:** one controlled test is not a universal shield; undiscovered objects, object structure and short-warning cases constrain response.

Verdict. India should value prestige missions for safety, science and industrial learning, while stating honestly that indigenous crewed flight and planetary-defence capability remain staged, not completed endpoints.

Historical Mains Core routes - station, observatory and lunar capability

- **India's space-station aspiration (2019):** a station extends a one-off crewed-flight capability into sustained human presence, microgravity research, docking/resupply, life-support and long-duration operations. **Evidence:** BAS-01 is approved with a target, while Gaganyaan's uncrewed/crewed steps remain staged. **Qualification:** a target module is not a functioning station; safety, budget, research use and launch/recovery maturity decide value.
- **James Webb Space Telescope (2022):** JWST is a large infrared space observatory operating around Sun-Earth L2; infrared observations help study faint/distant early-universe objects, star/planet formation and exoplanet atmospheres. **Qualification:** it is an international observatory, not an Indian launch vehicle or a generic "more powerful telescope" claim; its science benefit depends on instruments/data interpretation.
- **Chandrayaan-3 (2023):** the mission combined controlled lunar soft landing with in-situ surface science near the south polar region. **Significance:** it demonstrated landing/surface operations that support future mission sequencing. **Qualification:** it did not itself demonstrate sample return, human lunar flight or a permanent lunar base.

CLOSING RECALL FLOW - CORE SYNTHESIS - PLANETARY DEFENCE AND DATED MISSION AUDIT

```
START / CONCEPT: CORE SYNTHESIS - Planetary defence and dated mission audit
        |
        v
EXACT TERMS: CORE SYNTHESIS · Planetary defence · dated mission audit · Mechanism chain · Qualified
use · Subject
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Planetary-defence boundary - Volatile mission boundary
Qualified use: Close with detect-characterise-warn-deflect and fresh official evidence.
        |
        v
CONSEQUENCE / CONTRAST: Treat crewed flight, planetary science and planetary defence as adjacent
but different capability ladders: one test or foreign mission cannot certify all three.
        |
        v
UPSC TRAP / ANSWER-USE: Claim: Planetary defence is chiefly an early-warning problem -> NASA's DART
changed Dimorphos's orbit through a kinetic-impact demonstration -> it shows a deflection mechanism
can work with warning and characterization -> qualification: one controlled test is not a universal
shield; undiscovered objects, object structure and short-warning cases constrain response.
        |
        v
ANSWER-GRABBING FORMULATION: Read Planetary defence and dated mission audit as a sequence: identify
Planetary-defence boundary, follow the named mechanism to its user or output, and stop at the last
verified status rather than assuming the next stage.
```

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Human-rating boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: A. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Human-rating boundary?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: B. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Human-rating boundary without changing its institution, unit or status?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: C. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Human-rating boundary?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: D. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Orbital-module boundary?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: A. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Orbital-module boundary?

A. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. B. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

Answer: B. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Orbital-module boundary without changing its institution, unit or status?

A. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: C. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Orbital-module boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: D. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Crew-escape boundary?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

Answer: A. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Crew-escape boundary?

A. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: B. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Crew-escape boundary without changing its institution, unit or status?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: C. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Crew-escape boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: D. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Life-support boundary?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: A. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Life-support boundary?

A. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: B. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a

satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Life-support boundary without changing its institution, unit or status?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: C. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Life-support boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

Answer: D. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Qualification-ladder boundary?

A. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. B. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: A. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Qualification-ladder boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

Answer: B. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Qualification-ladder boundary without changing its institution, unit or status?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: C. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Qualification-ladder boundary?

A. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: D. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Air-drop-test boundary?

A. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: A. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Air-drop-test boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: B. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Air-drop-test boundary without changing its institution, unit or status?

A. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: C. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Air-drop-test boundary?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. D. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved.

Answer: D. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Uncrewed-crewed boundary?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. C. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. D. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

Answer: A. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Uncrewed-crewed boundary?

A. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. B. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: B. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Uncrewed-crewed boundary without changing its institution, unit or status?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: C. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Uncrewed-crewed boundary?

A. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: D. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an

achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Recovery-chain boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: A. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Recovery-chain boundary?

A. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. B. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: B. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Recovery-chain boundary without changing its institution, unit or status?

A. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: C. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Recovery-chain boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: D. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Axiom-Gaganyaan boundary?

A. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: A. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Axiom-Gaganyaan boundary?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: B. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Axiom-Gaganyaan boundary without changing its institution, unit or status?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

Answer: C. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Axiom-Gaganyaan boundary?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

Answer: D. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital

module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Mission-status boundary?

A. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: A. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Mission-status boundary?

A. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: B. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Mission-status boundary without changing its institution, unit or status?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. C. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. D. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.

Answer: C. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Mission-status boundary?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: D. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Chandrayaan-1 boundary?

A. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: A. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Chandrayaan-1 boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: B. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Chandrayaan-1 boundary without changing its institution, unit or status?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: C. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Chandrayaan-1 boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: D. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Chandrayaan-2 boundary?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: A. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Chandrayaan-2 boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: B. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Chandrayaan-2 boundary without changing its institution, unit or status?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: C. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Chandrayaan-2 boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: D. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Chandrayaan-3 boundary?

A. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

Answer: A. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Chandrayaan-3 boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: B. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Chandrayaan-3 boundary without changing its institution, unit or status?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: C. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Chandrayaan-3 boundary?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.

Answer: D. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Chandrayaan-4 boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: A. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Chandrayaan-4 boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: B. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Chandrayaan-4 boundary without changing its institution, unit or status?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: C. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Chandrayaan-4 boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

Answer: D. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Aditya-L1 boundary?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: A. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Aditya-L1 boundary?

A. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. B. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. C. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. D. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.

Answer: B. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Aditya-L1 boundary without changing its institution, unit or status?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

Answer: C. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Aditya-L1 boundary?

A. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. B. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: D. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Lagrange-station-keeping boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.

Answer: A. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Lagrange-station-keeping boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: B. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Lagrange-station-keeping boundary without changing its institution, unit or status?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: C. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Lagrange-station-keeping boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point.

Answer: D. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Mars-Venus boundary?

A. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: A. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Mars-Venus boundary?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: B. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Mars-Venus boundary without changing its institution, unit or status?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: C. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Mars-Venus boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. C. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: D. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Capability-ladder boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: A. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Capability-ladder boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: B. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Capability-ladder boundary without changing its institution, unit or status?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. C. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: C. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Capability-ladder boundary?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

Answer: D. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Planetary-defence boundary?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: A. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Planetary-defence boundary?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: B. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Planetary-defence boundary without changing its institution, unit or status?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: C. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Planetary-defence boundary?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.

Answer: D. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile mission boundary?

A. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: A. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile mission boundary?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: B. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile mission boundary without changing its institution, unit or status?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: C. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile mission boundary?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: D. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2019 space-station, 2022 JWST, 2023 Chandrayaan-3 and 2024 asteroid Mains demands plus RTG, LISA and microgravity objective concepts here. No answer key, telescope specification or mission outcome is invented.

PYQ DEMAND CARD 1 - 2019 GS-III

Demand: Discuss India's space-station plan and its benefits for the space programme.

Status: Verified routed Mains demand.

Model solution: Human-rating boundary: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Life-support boundary:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Qualification-ladder boundary:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Uncrewed-crewed boundary:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Recovery-chain boundary:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Axiom-Gaganyaan boundary:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2022 GS-III

Demand: Discuss the features, goals and human benefits of the James Webb Space Telescope.

Status: Verified routed Mains demand; this package carries the mission-class and scientific-observatory method, not invented telescope specifications.

Model solution: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2023 GS-III

Demand: Explain Chandrayaan-3 mission tasks, subsystems and the Virtual Launch Control Centre.

Status: Verified routed Mains demand; only owner-supported lunar capability and subsystem distinctions are used.

Model solution: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Chandrayaan-3 boundary:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Chandrayaan-4 boundary:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 4 - 2024 GS-III

Demand: Discuss asteroids, extinction threat and prevention strategies.

Status: Verified routed Mains demand.

Model solution: Planetary-defence boundary: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 5 - 2024-2025 Prelims GS-I

Demand: RTG and microgravity-mission concept family.

Status: Verified routed objective concepts; official keys are not reproduced or inferred.

Model solution: Axiom-Gaganyaan boundary: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Mission-status boundary:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why human rating is more than an ordinary launcher success record. Answer in about 150 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.
- Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.
- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish Axiom-4 experience from indigenous Gaganyaan capability. Answer in about 150 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.
- Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.
- Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse the Gaganyaan qualification ladder before crewed flight. Answer in about 250 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Air-drop-test boundary. **Named evidence/example:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Uncrewed-crewed boundary. **Named evidence/example:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.
- The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.
- Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.
- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.
- An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved.
- A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Air-drop-test boundary. **Named evidence/example:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Uncrewed-crewed boundary. **Named evidence/example:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Explain India's lunar capability progression without overstating sample-return readiness. Answer in about 250 words.

Model thesis: Claim: Chandrayaan-1 boundary. **Named evidence/example:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-2 boundary. **Named evidence/example:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-3 boundary. **Named evidence/example:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.
- Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.
- Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.
- Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.
- Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

Qualified conclusion: Claim: Chandrayaan-1 boundary. **Named evidence/example:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-2 boundary. **Named evidence/example:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-3 boundary. **Named evidence/example:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's human-spaceflight and planetary roadmap through parallel capability ladders. Answer in about 300 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Aditya-L1 boundary. **Named evidence/example:** Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mars-Venus boundary. **Named evidence/example:** Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; 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Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.
- Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.
- Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.
- Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.
- Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.
- Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.
- Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.
- Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Aditya-L1 boundary. **Named evidence/example:** Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mars-Venus boundary. **Named evidence/example:** Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; 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no later rung may be inferred from an earlier announcement or test. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a status-disciplined science and safety framework for crewed, planetary and asteroid missions. Answer in about 300 words.

Model thesis: Claim: Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lagrange-station-keeping boundary. **Named evidence/example:** A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Planetary-defence boundary. **Named evidence/example:** Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.
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OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

> **Subject:** Science & Technology | **Tier:** Advanced | **GS Paper:** GS-III + GS-II (international cooperation/governance) + Prelims. **Core area:** Human-spaceflight strategy, mission-status discipline, planetary roadmap and science-policy signalling. **Grounded in:** ISRO Gaganyaan page (https://www.isro.gov.in/Gaganyaan.html); ISRO Axiom-4 launch and conclusion notes (https://www.isro.gov.in/ISRO_EN/Successful_Launch_of_Axiom_Mission.html ; https://www.isro.gov.in/Axiom04_mission_successfully_concluded_return_ISRO_Gaganyatri_ShubhanshuShukla.html); Integrated Air Drop Tests and IMAT-05 (https://www.isro.gov.in/Integrated_Air_Drop_Test_for_Gaganyaan_Missions.html ; https://www.isro.gov.in/ISRO_conducts_Second_Integrated_Air_Drop_Test_for_Gaganyaan.html ; https://www.isro.gov.in/Integrated_Main_Parachute_Air_Drop_Test.html); Pad Abort Test note (https://pib.gov.in/PressReleasePage.aspx?PRID=1540066); ISRO Chandrayaan pages (https://www.isro.gov.in/Chandrayaan-1_science.html ; https://www.isro.gov.in/Chandrayaan2_science.html ; https://www.isro.gov.in/Chandrayaan3.html); ISRO Aditya-L1 pages (https://www.isro.gov.in/Aditya_L1.html ; https://www.isro.gov.in/ISRO_EN/halo-orbit-insertion-adtya-l1.html); Cabinet approvals of 18 Sep 2024 (https://pib.gov.in/PressReleasePage.aspx?PRID=2055978 ; https://pib.gov.in/PressReleasePage.aspx?PRID=2055989 ; https://www.pib.gov.in/PressReleasePage.aspx?PRID=2055982); Gaganyaan status replies (https://pib.gov.in/PressReleasePage.aspx?PRID=2205291 ; https://pib.gov.in/PressReleasePage.aspx?PRID=2227019) - re-verified 2 Aug 2026. FACT: = source-grounded | ANALYSIS: = analytical inference | CURRENT: = current/dated development. *Companion:* `basic/03_Human-Spaceflight-Gaganyaan-and-Planetary-Missions.md`.

1. Analytical frame

ANALYSIS: Human spaceflight is not "a bigger satellite launch." It converts a *mission-success* engineering culture into a *crew-survival* culture, where the governing metric is **probability of loss of crew**, not probability of mission success. That forces an entirely different qualification pyramid: propulsion and structural margins, a Crew Escape System usable from pad to upper atmosphere, environmental control and life support, parachute-based deceleration and recovery, abort modes at every flight phase, and human-rating of the launcher itself. Analytically, therefore, the correct ladder is *component test -> integrated ground test -> abort/air-drop test -> uncrewed orbital flight -> crewed flight -> sustained presence (space station)*. Planetary missions sit on a parallel but different ladder - orbit insertion, soft landing, sample return - where the binding constraints are deep-space navigation, autonomy under signal delay and mass budgets, not life support. Never merge the two ladders, and never read an approved programme as a flown one.

2. Visual foundation

```
CAPABILITY ESCALATOR
launch reliability -> human rating -> abort safety -> crew systems -> docking/station capability
                     -> soft landing -> sample return -> long-duration habitation

STATUS DISCIPLINE
completed != operational != approved != under development != concluded mission
```

Mission class	Advanced analytical cue
Gaganyaan	Human-spaceflight capability and systems integration
Bharatiya Antariksh Station	Long-duration presence and docking/station technologies
Chandrayaan-1/2/3/4	Progression from orbital science to landing to sample-return ambition
Aditya-L1	Scientific observatory logic beyond planetary landing glamour
MOM / Venus Orbiter	Comparative planetology and deep-space operations learning

3. Essential definitions

Concept	Exam-ready meaning
FACT: **Mission-status discipline**	Practice of distinguishing completed, operational, approved, under-development and concluded missions precisely.
FACT: **Crew escape system**	Abort-capability layer that moves the crew module to safety in case of emergency. Its validation ladder in India ran Pad Abort Test (2018) -> Test Vehicle mission TV-D1 (2023); a **Test Vehicle abort demonstration is not an orbital mission**.
ANALYSIS: **Human-rating vs mission success record**	Human-rating adds structural/propulsion margins, redundancy, vehicle health monitoring, benign failure modes and abort provisions. A launcher can have a strong success record and still not be human-rated; conversely human-rating cannot be inferred from consecutive successes.
ANALYSIS: **Probability of Loss of Crew (P(LOC))**	The governing safety metric for crewed vehicles, distinct from probability of mission success. It is why the qualification burden - and hence the timeline - for Gaganyaan is categorically different from a satellite launch campaign.
ANALYSIS: **Qualification test hierarchy**	Ground/structural test -> component qualification -> **air-drop test of parachute and recovery** (IADT/IMAT, no rocket) -> **abort demonstration** (Test Vehicle) -> **uncrewed orbital flight (G1)** -> crewed flight. Each rung answers a different failure question; skipping the vocabulary is a common answer error.
FACT: **Human-spaceflight ecosystem**	Launcher, life support, training, recovery, mission control and safety architecture taken together.
FACT: **Comparative planetology**	Studying similarities and differences across planets/moons to understand planetary evolution.
FACT: **Sample-return capability**	Ability not just to land but to retrieve extraterrestrial material and safely return it to Earth - requiring ascent from the target body, rendezvous/transfer or direct return, re-entry survival and planetary-protection containment.
ANALYSIS: **Station-keeping at L1**	A halo orbit around Sun-Earth L1 is only quasi-stable; the spacecraft must fire thrusters periodically, so **propellant budget, not instrument life, often sets mission duration** for L1 observatories such as Aditya-L1.
FACT: **Microgravity research platform**	Orbital environment used to study biological, material and physical processes in low gravity.

4. Mechanism / how it works

1. Human spaceflight is a systems-integration challenge: launcher reliability, crew escape, module habitability, astronaut training, mission control and sea recovery must all work together.
2. Planetary missions are science-plus-technology missions: orbit insertion, landing, surface operations, communication and data interpretation all matter.
3. A state typically climbs capability in stages: orbital observation first, then landing precision, then return/docking/habitation technologies.
4. International human-spaceflight exposure, such as Axiom-4/ISS participation, can compress learning in procedures and experiments, but cannot substitute for indigenous launch and recovery competence.
5. Therefore, advanced answers should interpret missions as part of a roadmap rather than as isolated headline events.

5. Institutions and programmes

Core institutions

- FACT: **HSFC/ISRO:** core human-spaceflight coordinating body for Gaganyaan-related activity.

- FACT: **Training and recovery network:** astronaut training, simulation, aero-medical support and naval recovery are all institutional parts of the programme.
- FACT: **Mission-science ecosystem:** lunar, solar and planetary missions draw on tracking, payload science, propulsion and mission-operations capabilities across ISRO centres.

Governance / strategic debates

- ANALYSIS: Human-spaceflight programmes are often criticised as prestige projects, but the deeper policy question is whether they generate durable systems capability.
- ANALYSIS: Space-station ambition raises governance questions of cost, research prioritisation, international collaboration and long-term operational sustainability.
- ANALYSIS: Planetary missions must be assessed not only by symbolism but by data return, technology maturation and downstream industrial learning.
- ANALYSIS: Mission sequencing matters: over-ambitious jumps can create schedule slippage, while overly conservative sequencing can reduce strategic momentum.

6. Indian applications, strategic significance and limitations

Strategic and scientific significance

- FACT: Gaganyaan creates autonomous human-spaceflight capability rather than dependence on external launchers for crewed access.
- FACT: BAS planning signals a shift from one-off crewed demonstration to sustained orbital presence ambition.
- FACT: Chandrayaan progression shows India moving from scientific discovery to technology demonstration and sample-return aspiration.
- FACT: Aditya-L1 broadens India’s space profile beyond the Moon/Mars narrative into solar and space-weather science.

Implementation constraints

- ANALYSIS: Human-spaceflight tolerates far less failure risk than uncrewed missions; schedule caution is a structural requirement, not mere hesitation.
- ANALYSIS: Station-building and sample-return missions need docking, long-duration life support, re-entry precision and logistics depth.
- ANALYSIS: Deep-space and crewed missions compete for budget, industrial bandwidth and testing windows.
- ANALYSIS: Public discourse often over-rewards spectacular “firsts” and under-values sustained mission-operations capability.

Safety / ethics / governance caution

- ANALYSIS: Crew safety, radiation risk, mission-abort reliability and recovery readiness are ethical, not merely technical, obligations.
- ANALYSIS: Human-spaceflight prestige should not crowd out transparent justification of scientific and developmental returns.

7. Must-Know Facts for Prelims

- FACT: Gaganyaan’s official concept is a three-member crew to about 400 km orbit for about three days with recovery in Indian sea waters.
- FACT: HLVM3 is the launcher identified for Gaganyaan.
- FACT: Chandrayaan-2 should be classified as orbiter success with lander soft-landing failure, not as total failure.
- FACT: Chandrayaan-3 achieved successful soft landing and surface science operations.
- FACT: Aditya-L1 is a solar observatory in halo orbit around Sun-Earth L1, not a planetary lander.
- FACT: Mangalyaan is historically successful but officially treated in recent references as a concluded mission.
- FACT: Venus Orbiter Mission and Chandrayaan-4 are approved future missions, not operational missions.

8. UPSC traps

- WRONG: Axiom-4 is India's indigenous human-spaceflight mission. -> It is an international mission involving an Indian astronaut; Gaganyaan remains India's indigenous crewed-capability project.
- WRONG: Chandrayaan-2 and Chandrayaan-3 differ only by chronology. -> Chandrayaan-3 is specifically associated with successful soft landing after Chandrayaan-2's lander setback.
- WRONG: Aditya-L1 is a "planetary mission." -> It is a solar observatory mission.
- WRONG: A space station announcement means the station already exists. -> Bharatiya Antariksh Station is an approved roadmap item, not an operational facility.
- WRONG: Mars legacy guarantees Venus success. -> Each destination imposes different scientific and engineering challenges.

9. CURRENT: Current anchor

- CURRENT: **06 Jan 2024 | Aditya-L1 - operational in halo orbit.** ISRO confirmed successful insertion at Sun-Earth L1.
- CURRENT: **18 Sep 2024 | BAS-1 and Gaganyaan follow-on scope - approved.** Cabinet approval of an expanded Gaganyaan scope including **BAS-01, targeted December 2028**.
- CURRENT: **18 Sep 2024 | Venus Orbiter Mission and Chandrayaan-4 - approved.** Chandrayaan-4 approved as lunar sample-return technology (36-month completion wording); Venus opportunity stated as **March 2028**.
- CURRENT: **25 Jun - 15 Jul 2025 | Axiom Mission 4 - flown and concluded.** Group Captain Shubhanshu Shukla flew as mission pilot on an 18-day mission with seven Indian experiments.
- CURRENT: **24 Aug 2025 / 10 Apr 2026 / 7 Jul 2026 | IADT-01, IADT-02 and IMAT-05 - conducted.** Air-drop and main-parachute qualification tests supporting the first uncrewed orbital mission G1.
- CURRENT: **17 Dec 2025 / 12 Feb 2026 | Gaganyaan - crewed flight targeted 2027-28.** Official material described continuing G1 module checks, integration and simulations; **no Gaganyaan orbital flight had taken place** as of the latest verified statement.

Verified current anchor	Analytical use in answers
Aditya-L1 operating at L1	Use to show India's space-science expansion beyond landing-centric narratives - and to make the point that L1 missions are propellant-budget-limited, not instrument-limited.
BAS-01 approved with a December 2028 target	Use to argue that human-spaceflight ambition is shifting from a one-off demonstration to *continuity of presence* - a far harder logistics, life-support and resupply problem.
Chandrayaan-4 and Venus approval	Use to show mission sequencing from landing success toward sample return and wider comparative planetology, while noting sample return requires ascent and rendezvous capabilities India has not yet demonstrated.
Gaganyaan's completed tests are abort and parachute-recovery tests; G1 has not flown; crewed flight targeted 2027-28.	Use as the model example of **status discipline**: a well-publicised test campaign is evidence of methodical safety qualification, not of achieved human spaceflight. Schedule slippage in a crewed programme should be read as safety governance working, not as failure.
Axiom-4 participation	Use to distinguish international operational exposure from indigenous capability-building - India bought *experience*, not *capability*.

ANALYSIS: **Currentness note:** The dated statuses above are accurate to the cited source date (latest re-verification 2 Aug 2026); verify later updates before exam use.

11. Mains framework / angles

- ANALYSIS: Use a roadmap frame: orbital science -> landing -> sample return -> human rating -> station capability.
- ANALYSIS: Distinguish science goals from strategic signalling but show that both can coexist in state-led space programmes.

- ANALYSIS: Bring out institutional learning and industrial capability, not just symbolic milestones.
- ANALYSIS: Conclude by emphasising that credible long-horizon space policy depends on status discipline, safety discipline and mission sequencing discipline.

> **Answer thesis:** India's contemporary space programme should be read as a sequenced capability architecture in which human-spaceflight, orbital science, soft-landing success, future sample-return plans and international mission exposure together indicate a shift from symbolic demonstration toward layered technological sovereignty - but only if mission statuses are described precisely and safety-science trade-offs are analysed honestly.

12. Probable questions

- ANALYSIS: **Prelims (practice):** Which of the following are correctly matched with their present status: Aditya-L1, Mangalyaan, Chandrayaan-4 and Bharatiya Antariksh Station?
- ANALYSIS: **Mains (10 marks, practice):** Why is mission-status discipline essential in writing answers on India's space programme? Answer in 150 words.
- ANALYSIS: **Mains (15 marks, practice):** Discuss the strategic and technological significance of India's shift from lunar orbiters and soft landers toward human-spaceflight, space-station planning and sample-return ambitions.

13. Study links

- FACT: Foundation companion: `basic/03_Human-Spaceflight-Gaganyaan-and-Planetary-Missions.md`.
- FACT: `01_Space-Programme-ISRO-Launch-Vehicles.md` - launcher base, human-rating and institutional context.
- FACT: `02_Satellites-NavIC-GAGAN-and-Applications.md` - underlying satellite and mission-support ecosystem.
- FACT: `01_Space-Programme-ISRO-Launch-Vehicles.md` - launch-infrastructure constraints (Third Launch Pad, NGLV) behind a sustained orbital presence.
- FACT: `16_Nanotechnology-and-Applications.md` - advanced systems, avionics and materials linkages.

CONSOLIDATED REGISTER NOTES

Human Spaceflight: Gaganyaan and Planetary Missions: SYSTEM, MISSION AND INSTITUTION MAP

1. **Human-rating boundary:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
2. **Orbital-module boundary:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.
3. **Crew-escape boundary:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.
4. **Life-support boundary:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.
5. **Qualification-ladder boundary:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.
6. **Air-drop-test boundary:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved.
7. **Uncrewed-crewed boundary:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.
8. **Recovery-chain boundary:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

9. **Axiom-Gaganyaan boundary:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.
10. **Mission-status boundary:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.
11. **Chandrayaan-1 boundary:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.
12. **Chandrayaan-2 boundary:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.
13. **Chandrayaan-3 boundary:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.
14. **Chandrayaan-4 boundary:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.
15. **Aditya-L1 boundary:** Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.
16. **Lagrange-station-keeping boundary:** A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point.
17. **Mars-Venus boundary:** Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.
18. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.
19. **Planetary-defence boundary:** Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.
20. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Human Spaceflight: Gaganyaan and Planetary Missions: CATEGORY, STATUS AND NUMBER FIREWALLS

- Do not infer human rating from launch reliability alone.
- Do not merge Crew Module and Service Module functions.
- Do not call an abort test an orbital mission.
- Do not omit life support from human-spaceflight architecture.
- Do not merge component, air-drop, abort and orbital tests.
- Do not call an air-drop test a rocket flight.
- Do not rewrite a target date as an achieved mission.
- Do not stop the human-spaceflight chain at launch.
- Do not call Axiom-4 an indigenous Gaganyaan flight.
- Do not merge approved, operational, completed and concluded.
- Do not equate lunar orbit science with soft landing.
- Do not call Chandrayaan-2 a total failure.
- Do not infer sample return from Chandrayaan-3.
- Do not treat Chandrayaan-4 approval as returned samples.
- Do not call Aditya-L1 a solar landing.

- Do not describe L1 as a completely stable fixed point.
- Do not infer Venus mission success from Mars legacy.
- Do not merge distinct mission-capability ladders.
- Do not describe one deflection test as full planetary defence.
- Do not invent crew, dates, payloads, findings or mission status.

Human Spaceflight: Gaganyaan and Planetary Missions: ANSWER-WRITING SPINE

```
CLASSIFY THE MISSION AND ITS CAPABILITY LADDER
-> DEFINE HUMAN RATING MODULES ESCAPE LIFE SUPPORT AND RECOVERY
-> ORDER COMPONENT AIR-DROP ABORT UNCREWED AND CREWED TESTS
-> SEPARATE INTERNATIONAL EXPOSURE FROM INDIGENOUS CAPABILITY
-> CLASSIFY LUNAR SOLAR MARS VENUS AND ASTEROID MISSIONS
-> DATE-STAMP APPROVAL TEST LAUNCH OPERATION AND COMPLETION
-> CONCLUDE ON SAFETY SCIENCE INDUSTRIAL LEARNING AND MATURITY
```

Human Spaceflight: Gaganyaan and Planetary Missions: LIVE-SOURCE AND VOLATILE-CLAIM BOUNDARY

The ISRO Gaganyaan page returned title-only on 2026-09-03. Repository owners remain the bounded source; crew, schedule, tests, payloads, findings, spacecraft health and mission status require a dated official update.

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

High-yield use: Follow the panels as one topic-specific conceptual atlas: root question, classifications, processes or arguments, comparisons, objections and a final answer-writing spine. This text-native master is separate from both graphical flowchart artifacts.

ASCII MASTER FLOW - PANEL 1/12: Human-rating pyramid

```
LAUNCH RELIABILITY -> necessary starting condition
MARGINS AND REDUNDANCY -> tolerate selected failures
HEALTH MONITORING -> detect unsafe vehicle conditions
ABORT MODES -> protect crew during ascent
HUMAN RATING -> system qualification, not a success count
```

ASCII MASTER FLOW - PANEL 2/12: Gaganyaan module map

```
CREW MODULE -> habitable and re-entry capable
SERVICE MODULE -> propulsion, power and support
ORBITAL MODULE -> Crew Module plus Service Module
LAUNCH VEHICLE -> ascent platform, separate from spacecraft
RULE -> module, spacecraft and launcher are different layers
```

ASCII MASTER FLOW - PANEL 3/12: Crew-survival chain

```
ECLSS -> breathable and thermally controlled cabin
CREW ESCAPE -> separation from a failing launcher
RE-ENTRY -> controlled atmospheric return
PARACHUTES -> staged deceleration
RECOVERY -> splashdown, extraction and medical support
```

ASCII MASTER FLOW - PANEL 4/12: Qualification ladder

```
COMPONENT / GROUND TEST -> verify subsystem behaviour
AIR-DROP TEST -> parachute and recovery sequence
TEST-VEHICLE ABORT -> escape under a selected flight condition
UNCREWED ORBITAL FLIGHT -> integrated mission without crew
CREWED FLIGHT -> later accepted human risk
```

ASCII MASTER FLOW - PANEL 5/12: International-exposure firewall

```
AXIOM-4 -> international ISS mission experience
FOREIGN LAUNCHER / SPACECRAFT -> external transport systems
INDIAN EXPERIMENTS -> operational and scientific exposure
GAGANYAAN -> indigenous launch, module and recovery objective
RULE -> experience complements but does not certify capability
```

ASCII MASTER FLOW - PANEL 6/12: Mission-status ladder

```
ANNOUNCED / PROPOSED -> intent
APPROVED -> authority and programme sanction
UNDER DEVELOPMENT / TESTED -> work and evidence rung
LAUNCHED / INSERTED / OPERATIONAL -> distinct flight milestones
COMPLETED / PARTIAL / FAILED / CONCLUDED -> outcome vocabulary
```

ASCII MASTER FLOW - PANEL 7/12: Lunar capability ladder

```
CHANDRAYAAN-1 -> orbital science and lunar-water evidence
CHANDRAYAAN-2 -> orbiter success, lander setback
CHANDRAYAAN-3 -> soft landing and surface operations
CHANDRAYAAN-4 -> approved sample-return technology mission
RULE -> landing does not prove ascent, docking or Earth return
```

ASCII MASTER FLOW - PANEL 8/12: Solar-observatory geometry

```
SUN-EARTH SYSTEM -> defines Lagrange-point geometry
L1 REGION -> continuous solar-view advantage
HALO ORBIT -> three-dimensional path around L1
STATION KEEPING -> periodic correction and propellant use
ADITYA-L1 -> solar observatory, not a landing mission
```

ASCII MASTER FLOW - PANEL 9/12: Planetary-status matrix

```
MARS ORBITER MISSION -> historic mission, now concluded
VENUS ORBITER MISSION -> approved future mission in owner
LUNAR SAMPLE RETURN -> ascent and return architecture
SOLAR OBSERVATORY -> remote sensing from L1 halo orbit
RULE -> destination and status both control the description
```

ASCII MASTER FLOW - PANEL 10/12: Parallel capability ladders

```
HUMAN -> abort, life support, uncrewed, crewed, station
LUNAR -> orbit, soft land, rove, sample return
SOLAR -> transit, halo insertion, station keeping, observation
PLANETARY -> cruise, insertion, science operations
RULE -> one ladder cannot certify another
```

ASCII MASTER FLOW - PANEL 11/12: Planetary-defence rail

```
DETECTION -> discover potentially hazardous objects
CHARACTERISATION -> refine orbit, size and composition
WARNING TIME -> determines response feasibility
DEFLECTION -> kinetic impact is one demonstrated concept
RESIDUAL RISK -> one test is not a complete shield
```

ASCII MASTER FLOW - PANEL 12/12: Mission answer spine

```
CLASSIFY -> human lunar solar planetary or defence mission
TRACE -> minimum technical and safety chain
NAME -> one verified achievement or approved next rung
SEPARATE -> test launch operation completion and outcome
VERIFY -> dated crew schedule payload finding and status source
```